Behavioral Effects of Peer Separation, Isolation, and Reunion on Adolescent Male Rhesus Monkeys

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Six adolescent male rhesus monkeys which had lived in pairs for 6-8 months were studied for behavioral adaptation to 2 weeks of separation-isolation and 4 days of reunion. The response to separation was 1-3 days of increased locomotion and object-oriented behavior followed by a long term increase in self-directed behaviors. The latter phase differed significantly from the anaclitic depression of infants separated from their mothers. Reunion produced one day of intense social activity followed by a return to behavior patterns characteristic of the preseparation period. A large component of the initial motor excitement following separation may be nonspecific response to gross change in the environment.

The disruption of close personal relationships is a major source of emotional stress in human beings. Like humans, the nonhuman primates form close social relationships, thus offering the opportunity to study the formation and disruption of emotional bonds in a controlled fashion. Systematic studies in this area are a relatively recent development in the animal behavior literature, and as pointed out in a recent review (McKinney & Bunney, 1969), the results have been encouraging.

To date, most of the studies carried out in macaques have dealt with the mother-infant relationship (Kaufman & Rosenblum, 1967; Seay & Harlow, 1965; Seay, Hansen, & Harlow, 1962; Jensen, 1968; Hinde, Spencer-Booth, & Bruce, 1966). Such studies have demonstrated a behavioral syndrome in infant monkeys quite similar to the anaclitic depression of human infants (Spitz, 1946). This experimental model of bond disruption has limitations, however. One practical limitation in carrying the studies to a physiological or biochemical level is the considerable expense of mother-infant pairs; a theoretical limitation may lie in the nature of the response to disruption of intense social relationships as related to the age of the individuals

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involved. Thus, from research-economics and theoretical points of view other kinds of interpersonal relationships than mother-infant pairs must be explored.

As the strength of the maternal bond decreases, the young monkey's major relationships come to be with other monkeys of the same sex and age group. This attachment, as judged by frequencies of rough and tumble play and "incipient sexual behavior," peaks in the 2nd year of life and remains intense into adolescence (Hinde & Spencer-Booth, 1967; Hansen, 1966). The purpose of the present study was to explore the effects of manipulating the intense peer relationship seen in young male macaques.

A pilot study involving 2 pairs of adolescent monkeys who had lived together for many months was carried out to establish an effective experimental design. The parameters of the video sampling technique, the duration of the separation and reunion periods, and the mode of subdividing the separation and reunion periods into functionally significant phases were worked out in that study. The present report describes the responses of 6 further subjects to disruption and reestablishment of the adolescent peer bond.

Method

Subjects

The subjects were 6 adolescent male *Macaca mulatta.* Their ages at the time of separation were estimated on the basis of weight, sitting height, foot length, and dentition to be 31-34 months for 5 of the subjects and 48 months for the 6th; at the end of the study, the testes had not fully descended in any but the latter. All had been captured in the wild after the first year of life and were housed in small individual cages for 3 months quarantine before coming to the laboratory. In the laboratory interpersonal bonds were established by housing them in 3 pairs for 6 to 8 months before the experiment began.

Apparatus

The glass-fronted observation cage consisted of 3 compartments, each 80 X 50 X 150 cm, with perches at 3 levels (Fig. 1). The cage stood in a sound attenuated room. Vocal communication between animals in the different compartments was masked by noise (78 dB in the cage) from a white noise generator. All recording apparatus was located outside the observation room. Observations were made from video recordings obtained by a Sony Video recorder Model PV120 which was controlled by a system of clocks and relay logic modules to provide short recordings at desired intervals. On play-back, codes for free behavior were punched onto computer cards from which frequency histograms for all behaviors from all animals were prepared, displayed, and photographed on an oscilloscope attached to a Packard Bell 440 computer. An electronic clock was used to determine the amount of time in each recording sample given to locomotion. Continuous activity measures were obtained from ultrasonic motion detectors (Alton, Model SnM5322), the emitters and transducers of which were located in the ceiling of each compartment.

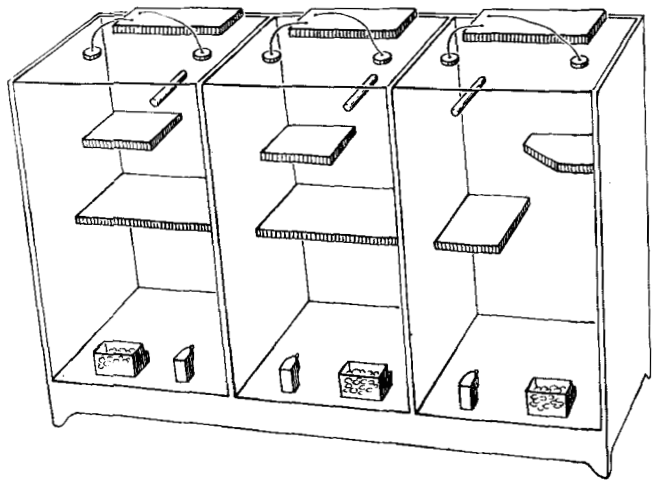


Fig. 1. Observation cage consisting of 3 compartments with ultrasonic motion detector (2 units) in the ceiling of each.

Procedure

Each pair was transferred from the home cage 6-8 weeks before separation, housed for at least 2 weeks in each of the end compartments of the observation cage and then transferred to the middle compartment. The experiment began after the pair had lived for at least 2 weeks in the middle compartment. Data were collected for 5 days in the "preseparation" condition (pair in middle compartment), 14 days in the "separation" condition (one subject in each end compartment), and 5 days in the "reunion" condition (pair in middle compartment). Recording difficulties reduced the data on reunion to 4 days.

Figure 2 illustrates the daily schedule followed with all pairs under all conditions. Laboratory personnel entered the room for cage cleaning and feeding for as short a time as possible during a 1-hr period in the middle of the day. Observations were omitted for an hour before and an hour after cage cleaning and feeding to minimize contamination of the data by behavior anticipating or evoked by these procedures.

Observations of free behavior were taken in 15-sec "time samples" at 8-min intervals for 3.3 hr beginning with light-on in the morning and for another 3.3 hr ending with light-off in the evening to give a total of 48 samples per day. In addition to the 15-sec time samples, the first 1/2-hr of reunion was recorded in its entirety.

The videotapes were analyzed for 54 different "behavioral items" (BI's) from a behavior catalogue constructed to cover all behaviors which the pilot study had shown could be reliably recorded by different observers. Each BI was defined in terms of unambiguous, visible criteria as a physical event without reference to prior concepts of motivation or social significance. Every time sample was observed independently by 2 investigators (D.B., W.M), each of whom punched the codes for his observations directly onto computer cards. The code for a BI was punched only once for a given animal in a given sample, no matter how many times it occurred. A BI was admitted to

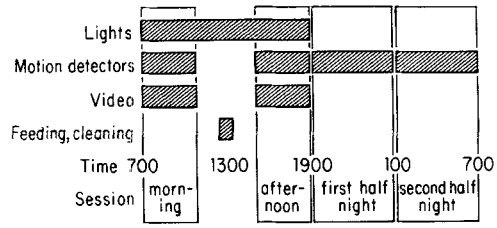


Fig. 2. Daily schedule. Recording sessions are indicated in lower portion.

the data if either observer noted it. In order to estimate the error incurred by this procedure, approximately 5% of the video samples were reviewed by the observers together to determine the proportion of recorded BI's which failed to meet the established definitions. This joint review revealed that less than 5% of the BI's admitted into the data were erroneous.[1]

Locomotion time was measured for each animal in every sample by viewing the videotapes again. The observer activated the electronic counter whenever the animal was moving and, at the end of the sample, recorded the time accumulated in seconds. The animal's "total locomotion time" for the day was the sum of locomotion times across samples. "Average locomotion time" was total locomotion time divided by the number of samples in which locomotion occurred.

Diurnal activity was monitored in 4 periods: the first 3.5 hr after lights-on in the morning, the last 3.5 hr before lights-off in the afternoon, the first 6 hr of night, and the last 6 hr of night. The score for each period was the proportion of the total activity recorded in the 4 periods occurring during that period. Analysis of these data was directed at determining whether the proportion of the total day's activity registered in a particular period changed during different phases of the experiment.

Food intake was monitored by daily weighing of the chow left from the previous day's feeding. The diet consisted of one-quarter apple per animal, Purina monkey chow *ad libitum,* water *ad libitum* and Isoniazid (20 mg per day) for tuberculosis prophylaxis.

Data Analysis

Data analysis consisted of 2 steps. First the frequencies of the BI's were scanned for similarities in patterns of change which would provide a first approximation for meaningful groupings of the BI's into categories. Then the frequencies of these behavior categories (BC's) and other variables such as food intake, locomotion time, etc. were examined for systematic changes in different phases of separation and reunion.

The 54 BI's were assigned to 21 behavior categories which were derived empirically in the following manner. A frequency histogram was plotted for each animal's performance of each BI across the 23 days of the experiment. The histograms for a given animal were sorted on the basis of general shape. To a large extent, the different profiles represented groups of BI's related to one another in some way. For instance, BI's such as "handles food," "carries food," "ingests," "chews," and "clears pouches" shared a profile which differed from the profile shared by BI's such as "looks out," "reaches out," "looks into floor pan of neighboring compartment," etc. The groups of

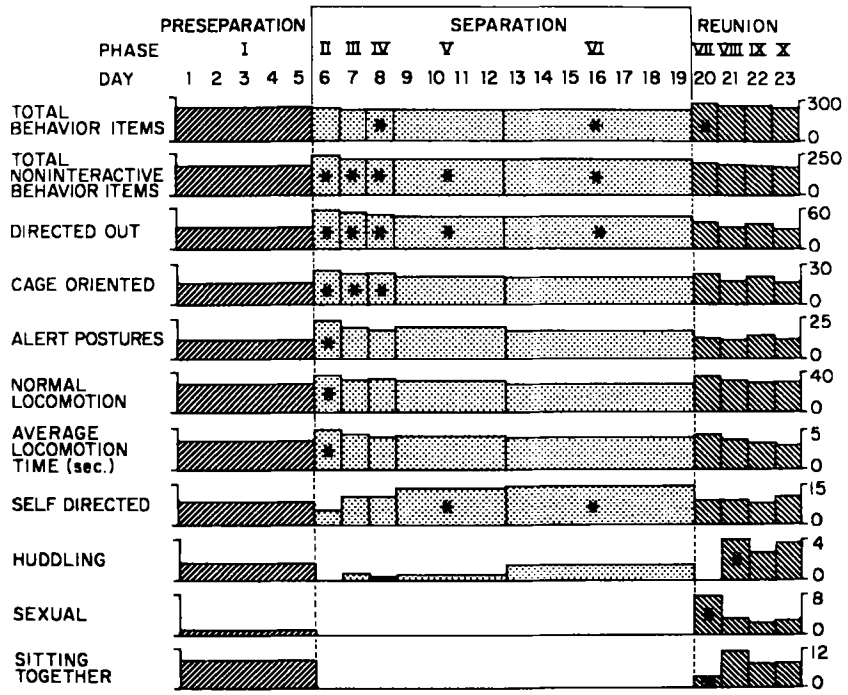


Fig. 3. Behavioral descriptors showing significant differences from preseparation levels during separation and/or reunion by Friedman's analysis of variance. Ordinate: mean value per animal per day; abscissa: phase of experiment.

*Difference from preseparation phase significant at $p \leq .05$ level by Dunnett's t statistic.

BI's based on profile were subdivided further into categories of 1 to 8 BI's each to reflect functionally related categories such as behavior directed out of cage, cage oriented behaviors, alert postures, etc. Every BI which was observed more than 4 times in any animal was assigned to a BC; no BI was assigned to more than one BC. The observed value of a given BC on a given day for a given animal was the sum of the frequencies of the BI's comprising that BC. Results of the pilot study had indicated that the major effects of separation and reunion could be identified by an analysis which considered the total experiment as comprising 10 phases (Fig. 3). Phase I consisted of the preseparation period; phases II, III, and IV of the 1st, 2nd, and 3rd days, respectively, of separation; phase V of the 4th through 7th days of separation; phase VI of the last 7 days of separation. Phases VII, VIII, IX, and X consisted of the 1st, 2nd, 3rd, and 4th days of runion.

Statistical tests were run on 28 "descriptors" to identify significant effects of different phases of separation and reunion. Friedman's analysis of variance (Siegel, 1956) was used to test whether each descriptor was influenced by the different phases of the experiment. Descriptors which showed a significant phase effect were then subjected to a parametric one-way fixed effects analysis of variance for repeated measures, and those for which $p \leq .05$ by this analysis were subjected to Dunnett's t statistic for comparing multiple treatments with a control (Winer, 1962).[2] Here the separation and reunion phases (phases II-X) were compared with the preseparation phase (phase I). Descriptors measured on an interval scale (food intake, total locomo-

tion, mean locomotion per frame) were subjected only to the parametric analysis of variance and Dunnett's t statistic with the significance criterion: $p \leqslant .05$, two-tailed. For most tests N = 6 for the number of individual animals on which observations were made. The methods of measuring food consumption and diurnal activity did not allow independent identification of the values for individual animals during phases when they were housed in pairs. Thus, for tests of significance on these descriptors N = 3, the number of pairs.

Descriptors which involved interaction between monkeys could not occur during the separation phases. Thus, all tests of significance for these descriptors compared only 5 phases, phase I with phases VII through X.

Changes in the distribution of activity within the diurnal cycle were tested for by carrying out Friedman's analysis of variance on each of the 4 recording periods separately. Thus, for instance, it was possible to judge whether the proportion of total activity registered in the late afternoon varied significantly from one phase of the experiment to another.

Results

Figure 3 shows the descriptors for which Friedman's analysis indicated a significant phase effect and for which Dunnett's statistic indicated the significant differences to be between preseparation and certain phases of separation and reunion.

Separation

The total number of BI's observed per day decreased significantly during separation. The fact that the number of noninteractive behaviors significantly increased during this period suggests that the decrease in total BI's was primarily attributable to the absence of another animal with whom to interact.

During the early separation period the increase in noninteractive BI's was attributable largely to increases in the frequencies of behavior directed outside the cage, cage-oriented behavior, alert postures, and normal locomotion.[3] The increase in locomotor activity was also reflected in increased locomotion time per sample. The behavior directed outside the cage persisted at a significantly elevated level throughout the separation period; behavior in the other categories dropped back to preseparation levels within 3 days. Self-directed behaviors increased steadily during the separation period.

The data were scanned for statistically nonsignificant descriptors in which the animals which did change showed a consistent pattern. Only one such descriptor emerged, viz., rocking. Two animals accounted for 95% of the rocking, and in both the changes in frequency paralleled those of self-directed behavior, i.e., rocking was rare in the first days of separation and increased in frequency in later phases.

Reunion

The first 1/2-hr of reunion was remarkable for the variety and intensity of 3 forms of behavioral interactions: sexual, aggressive, and grooming. In every pair, sexual be-

havior predominated for the first 2-5 min and remained quite intense for the first 15-30 min; every pair also showed a prolonged bout of aggressive behavior at some point during the first 1/2-hr. Again, the aggressive interchanges were bilaterally initiated with the members of the pair alternating between chasing and being chased. Such interaction appeared playful at the outset. In pair I, however, it became more intense after several minutes and the bout ended with several seconds of intense fighting. Two of the pairs also showed long bouts of grooming. All of these behaviors had been seen at one time or another during preseparation, but never in such intensity, such rapid alternation, and for such prolonged duration.

The adaptation to reunion as represented by the daily frequencies of different BC's was extremely rapid. As seen in Figure 3, all of the noninteractive BC's immediately returned to levels not significantly different from their preseparation levels. The 1st day continued to show a significantly elevated frequency of sexual behavior, and total locomotion tended to remain elevated. Less active forms of social interaction, including sitting together, embracing and sniffing and touching one another, were decreased in frequency compared to preseparation levels. Most behavior categories returned to preseparation levels by the 2nd day, and huddling was significantly increased on that day.

Altogether, 40% of the descriptors showed changes which were consistent from animal to animal in separation and reunion. The animals' individual behavior frequencies indicated (a) that each of the starred columns of the histograms in Figure 3 represented a difference from the preseparation value for at least 4 of the animals, (b) that those showing changes all changed in the same direction as the shift in the mean, and (c) that no consistency existed with regard to which animals failed to change from one descriptor to another. In particular, the few cases where individual animals failed to show the differences which reached statistical significance for the group as a whole could not be attributed to one of the pairs or to the one subject who was older than the others.

Of those descriptors which did not show significant changes, several merit comment because they represent variables which both the clinical and the experimental literature suggest are sensitive indicators of social stress. Change in appetite as reflected in frequency of food-water directed behaviors and in amount of chow consumed changed markedly in several subjects during early phases of separation: in some it increased, in others it decreased, but the net effect for the 6 animals as a group was no change. Stereotyped locomotion (pacing and jumping in place) was seen in 4 subjects, but, as with appetite, the pattern of change from phase to phase was not consistent. The diurnal distribution of gross activity showed no marked change in any subject.

Discussion

Under the conditions of this experiment, separation of adolescent rhesus monkeys which had lived together for a long period of time elicited a 2-stage response. In the first, the excitement stage, alertness, locomotor activity, and exploratory activity

showed an immediate increase. In the second stage these behaviors declined to preseparation levels; self-directed activities and, in some subjects, rocking gradually increased and remained at a plateau for the remainder of the separation period.

Reunion also elicited a 2-phase response: a very short phase of intense social activity was followed by a return of virtually all descriptors to preseparation levels.

The construction of behavioral categories by sorting profiles was utilized in this study to maximize the likelihood of detecting changes in groups of behaviors which might not correspond to classes one would *a priori* group together. Changes in unforeseen groupings frequently occur when one studies the effects of psychoactive drugs or selective brain lesions on behavior and might have occurred when, as here, the independent variable was a manipulation of the social environment. However, the categories generated in this study matched fairly closely the categories one would have postulated on the basis of motivational and ecological concepts. Since the categories were generated *post hoc* one cannot say that all of the changes starred in Figure 3 represent real changes. One can, however, conclude that if separation and reunion have any effects on the free behavior of adolescent monkeys under conditions represented by this experiment, they are of the nature described here. Sorting the BI's into other categories on whatever basis would not in all likelihood reveal patterns of change not reducible to those represented in Figure 3.

Comparing these results with results of other studies, one finds that the initial stage of the response to separation was quite similar to that seen in separation of mother-infant pairs. All such studies have described an initial stage of motor excitement lasting for 1 to 3 days (Kaufman & Rosenblum, 1967; Seay & Harlow, 1965; Seay *et al.*, 1962; Hinde *et al.*, 1966; Jensen & Tolman, 1962; Rowell & Hinde, 1963; Mason & Green, 1962). Bowlby (1960), who reported the same phenomenon in young children, has referred to it as the "protest" phase. In general the motor excitement has been ascribed to the stressfulness of separation from the mother or group.

Interestingly, in the present study, the first day of reunion was also characterized by excitement. Sexual behavior was significantly increased; frequency of locomotion tended to increase in a manner similar to that seen on the first day of separation. Sitting quietly together and huddling were significantly less frequent than in the preseparation period. This finding suggests that the motor excitement which characterizes the first stage of the response to separation should be looked upon as a general response to drastic change in the environment rather than as a specific response to the separation. In this age group, at least, a large change in the environment, whether it be subtraction or addition of another individual, had a generalized activating effect which appeared to take different forms of excitement, depending upon the nature of the new environment.

In further comparing the results of this study with those of studies in which younger animals were separated from their mothers, one finds that the second stage was quite different. Infants separated from their mothers pass into a syndrome characterized by huddling posture, dejected facies, and reduced locomotion, play, and exploratory behavior (Kaufman & Rosenblum, 1967; Seay & Harlow, 1965; Seay *et al.*, 1962; Hinde *et al.*, 1966). This stage has also been reported in the human literature. Spitz (1946) named it "anaclitic depression"; Bowlby (1960) refers to it as the "despair" phase of the response to loss of the mother.

Separation of adolescent peers in this study did not produce any signs of such a despair stage, either in the group as a whole or in individual animals. Activity levels did not drop and the huddling posture was seen, if anything, less frequently than in the preseparation phase. Behavior categories reflecting interest in the environment never fell below their preseparation frequencies.

In the mother-infant studies where separation lasted more than a week, the signs of "despair" were most pronounced in the first week. As these signs waned, self-directed oral behaviors (sucking and mouthing the fingers, toes, and limbs) and exploratory behavior rose to levels higher than in the preseparation periods (Seay *et al.*, 1962; Kaufman & Rosenblum, 1967). Such changes are in the direction of the second stage of adaptation seen in the present study and suggest that the ultimate adaptation to separation may not differ greatly in infants separated from their mothers vs adolescent males from familiar peers. The major difference may be that in reaching the final state the infants pass through a transient phase of hypoactivity which the peer-separated animals do not show.

Two major factors which one must consider in attempting to account for the different second phase effects of mother-infant separation and peer separation as carried out in this study are (1) previous separation experience and (2) age at the time of separation. Although the animals in this study had lived together for at least 25% of their lives, all must have been subjected to severe separation trauma when they were captured in the first year of life, and all had lived in individual cages for 3 months quarantine before their relationship was established in the laboratory. A recurrent theme in the psychiatric literature is that object loss in infancy or early childhood predisposes to depressive reactions in later life (Hill & Price, 1967). Were this the case, one would have expected our subjects' past experience to have made them more vulnerable to depression than animals not subjected to previous separation trauma. Yet they showed no behavioral signs of depression whatever.

Age is very likely the more important factor. Spitz (1946) defined anaclitic depression as an age-specific response of human infants in the second half of the first year of life; Bowlby (1960) observed the protest-despair syndrome in young children, 1 to 4 years of age. Many clinicians recognize that in older children and adolescents, statements expressing hopelessness, helplessness, and worthlessness (the "primary signs" of depression in psychiatric terminology) frequently accompany a quite different behavioral syndrome from that seen in anaclitic depression or adult depression. In this age group, "depressive thoughts and affect" are commonly associated with hyperactivity, delinquent behavior, "acting out," and the like (Glaser, 1967). Thus the behavior shown by adolescent monkeys separated from their peers in this study is not inconsistent with psychiatric descriptions of behavioral concomitants of adolescent depression in humans.

Notes

[1]Complete definitions of BI's and BC's and interobserver reliability scores of individual BI's available from authors on request.

[2]This statistical approach was chosen because there exists no non-parametric test for comparing multiple treatments with a control.

[3]BC's showing significant changes comprised the following BI's: directed outside cage (looks out, looks into floor pan, reaches out); cage directed (picks at cage, bites cage); alert postures (stands on hind legs, crouches); normal locomotion (walks, climbs); self-directed (scratches, grooms self, grooms genitals).

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